Viridis Research — certification record

Run `runs/2026-10-30_P2` 2026-10-30 project P2, task P2-T002 engine `fd227d5c3af7`
GitHub run: `https://github.com/jdhart81/viridis-engine/actions/runs/41068705066`

Sealed Statements for the Restoration, Afforestation and Harmonization Kernels: Faithful Lean Restatements of the Canon Formal Cores
Justin Hart, Viridis LLC

Labels awarded by the verifier

Claim	Label	Claim text (as submitted)
C1	**ASSUMED**	conjecture: Observation from a private local check, not a sealed claim: P2.Restoration, P2.Afforestation and P2.Harmonization, as proposed, are each provable in Lean v4.28.0 with pinned Mathlib using only propext, Classical.choice and Quot.sound, by porting the canon proofs.
C2	**ASSUMED**	conjecture: Observation, not a sealed claim: in Lean's division convention, the positivity hypotheses of the cubic efficiency law and the nonzero hypotheses of the alignment bound in P2.Afforestation are not needed for the conclusions; they are kept to match the canon and the kernel's parameter domain.

Other checks

- Lean toolchain leanprover/lean4:v4.28.0, Mathlib 8f9d9cff6bd7; kernel replay by `leanchecker`.
- References: 5 entries, 5 DOIs, 5 resolved, 0 unresolved, 0 unchecked.
- Agent paper: compiled.
- Statement proposals: 3.
- P2.Restoration: compiled; no probe tactic closes it; submitted for sealing
- P2.Afforestation: compiled; no probe tactic closes it; submitted for sealing
- P2.Harmonization: compiled; no probe tactic closes it; submitted for sealing

How to read this record

ASSUMED: not verified (the default). **PROVEN**: a Lean 4 proof of a statement sealed before the attempt, compiled with the pinned toolchain, matched definitionally to the sealed statement, using only the axioms propext, Classical.choice and Quot.sound, and replayed through the kernel. **TESTED/REPLICATED**: reproduced by the verifier (not yet enabled). **ENDORSED/ADOPTED**: granted by people. Novelty is never a label. **TRIVIAL** marks a sealed statement that one of the engine's probe tactics already closes.

This paper was drafted by an AI research agent (Claude, by Anthropic) running the Viridis research engine; formal proof search may use Aristotle (Harmonic). Justin Hart directs the research program. The labels on this page were assigned by an automated verifier, not by the agent, and only they carry certification. Claims in the paper that are not listed here as PROVEN are ASSUMED.

Reproduce everything from `REPRODUCE.md` in the run folder; the run's files are listed with hashes in `MANIFEST.sha256`.

Re-certifying the Viridis OS Decision Kernels: Carbon Continuity, and Sealed Statement Proposals for Restoration, Afforestation and Harmonization

Justin Hart
Viridis LLC

30 October 2026

Abstract

Viridis OS runs decision kernels whose formal cores were checked by tools that can no longer be re-run. This paper documents the project that brings each kernel to a re-checkable standard: the kernel's formal core is restated as a sealed, self-contained Lean 4 statement and re-proved under a pinned toolchain. The first kernel, *Carbon Continuity*, has a Lean proof of its sealed statement: a two-pool carbon portfolio can be kept from shrinking exactly when $(1-a)(1-d) \le rp$. This revision adds proposed sealed statements for the next three kernels. `P2.Restoration` restates the forest-nucleation core. The critical radius $r^\star = 2a\sigma/(3\Delta\mu)$ is positive and is the unique maximiser of the free energy $G(r) = -\Delta\mu r^3 + a\sigma r^2$ on $r \ge 0$, and the barrier satisfies $G(r^\star) = \frac{1}{2}\Delta\mu(r^\star)^3$. `P2.Afforestation` restates two results: seeding efficiency scales as the cube of the ecological drive, and the seeding alignment $\eta = \langle n, m\rangle^2/(\|n\|^2\|m\|^2)$ lies in $[0, 1]$. `P2.Harmonization` restates two properties of a scalar-price welfare theorem for N quadratic stewards: market clearing, and the absence of a duality gap. Each proposal copies the canon definitions and hypotheses verbatim and records the hash of the canon file. It also includes sanity examples with concrete numbers, among them negative controls. We point out where canon theorem names promise more than the theorems prove. We also point out which canon hypotheses are kept for faithfulness although the proofs do not need them. Sealing the proposals is a human decision, and no proof of any proposal is claimed here.

1 Introduction

Viridis OS runs a set of decision kernels. Each one is a small piece of software that turns ecological parameters into a decision. A kernel's rule is only as trustworthy as the mathematics behind it. The archived canon [3] proved those rules in Lean, but the certificates came from tooling that has since been retired. The project described here moves each kernel's assurance onto a re-checkable footing in two steps. First, a *sealed statement* fixes the exact proposition a kernel relies on. Second, a Lean proof of that statement is checked under Lean v4.28.0 with a pinned Mathlib [5].

The two steps are separate on purpose. Writing the statement is where most of the risk lies. A statement can be correct and easy to prove and still be the wrong one. It can be weaker than what the kernel uses, stronger than what the canon proved, or vacuous. The present revision is about that first step for three kernels: Restoration, Afforestation and Harmonization. The question is narrow. We ask what each kernel's rule *mathematically* relies on, and how to write that down self-contained and faithful to the canon theorem it replaces. We do not ask whether the models describe any real landscape.

2 Background and related work

Carbon Continuity. One disturbance-recovery cycle maps $(L, D) \mapsto (aL + rD,\ pL + dD)$. This is the nonnegative matrix $M = \begin{pmatrix} a & r \\ p & d \end{pmatrix}$, and the threshold theorem is a two-dimensional instance of the Collatz–Wielandt characterisation in Perron–Frobenius theory [1].

Restoration and Afforestation. The canon models a planted cluster as a nucleus of the forest phase inside a metastable degraded phase. It uses the free-energy form of classical nucleation theory [4]: a bulk gain $-\Delta\mu\, n$ competes with a surface cost $a\sigma\, n^{2/3}$. In the cube-root variable $r = n^{1/3}$ this becomes the cubic $G(r) = -\Delta\mu r^3 + a\sigma r^2$. The textbook facts are a critical size at the barrier top and a barrier equal to half the bulk term at the critical size. These are the mathematical content of `P2.Restoration`. The Afforestation efficiency $e = C/n^\star$ inherits the cube from $n^\star = (r^\star)^3$. The alignment score is a squared cosine, bounded by Cauchy–Schwarz.

Harmonization. The scalar-price theorem is the separable quadratic case of Lagrangian duality with one linear coupling constraint [2]. At the clearing price, the decentralised best responses solve the centralised problem, and there is no duality gap.

We claim no novelty for any of this mathematics. The contribution is a set of faithful, machine-checkable targets tied to deployed kernels.

3 Carbon Continuity (P2-T001)

The sealed module defines stepLiving$(a, r, L, D) = aL + rD$ and stepDurable$(d, p, L, D) = pL + dD$. Sustainable(a, d, r, p) means that some $L, D > 0$ satisfy $L \le aL + rD$ and $D \le pL + dD$.

Theorem 1 (`P2.CarbonContinuity`)**.** *For all real a, d, r, p with $0 \le a < 1$, $0 \le d < 1$, $r > 0$ and $p > 0$:* Sustainable$(a, d, r, p) \iff (1 - a)(1 - d) \le rp$.

Proof sketch. Sufficiency uses the witness $(L, D) = (r, 1-a)$. The living step is then an identity, and the durable step reduces to $rp - (1 - a)(1 - d) \ge 0$. For necessity, rearrange the two inequalities to $(1-a)L \le rD$ and $(1-d)D \le pL$, multiply them (both sides are positive), and cancel $LD > 0$. □

The previous revision gave a Lean proof of this statement (file `proofs/P2.CarbonContinuity.lean` of run 2026-10-29), with axioms `propext`, `Classical.choice` and `Quot.sound`.

```
theorem main : ViridisStatements.P2.CarbonContinuity.Statement := by
  intro a d r p _ ha1 _ hd1 hr _
  exact ⟨necessary a d r p ha1 hd1, sufficient a d r p ha1 hr⟩
```

We observe that the helper lemmas use only $a < 1$, $d < 1$ and $r > 0$.

4 Method for writing a sealed statement

For each kernel we followed the same procedure.

1. **Identify the canon theorems.** We take the theorems named in the task brief and check the canon file's SHA-256 against the pinned manifest (canon commit `80d32904`).

2. **Copy definitions verbatim.** We copy only the definitions the theorems use, with the same argument order, and drop the rest. Any added abbreviation must unfold to the canon expression exactly.

3. **Copy hypotheses verbatim.** Each conjunct of `Statement` keeps exactly the hypotheses of its canon theorem. We drop no hypothesis, even one that is mathematically unnecessary, because dropping it would strengthen the canon statement. We add none, because adding one would weaken it.

4. **State the conclusion as proved, not as named.** Where a canon docstring or name claims more than the Lean conclusion, the statement follows the conclusion.

5. **Pin the meaning.** We add sanity `example`s with concrete numbers, closed by `norm_num` or `simp` (`decide` does not apply to $\mathbb{R}$). They include negative controls showing that the key hypotheses or factors cannot be dropped.

6. **Test non-triviality.** We compile each module and probe `Statement` with 28 one-shot tactics after `unfold Statement`. The tactics include `rfl`, `decide`, `simp`, `norm_num`, `aesop`, `linarith`, `nlinarith`, `positivity`, `grind`, `field_simp`, `ring` and `polyrith`.

7. **Test feasibility.** Privately, we check that a port of the canon proof closes the proposed statement. This guards against proposing a false target. It is not a claim of proof.

5 The three proposals

5.1 `P2.Restoration`: the critical-nucleus go/no-go

Definitions: $r^\star(\Delta\mu, a, \sigma) = 2a\sigma/(3\Delta\mu)$ and $G(\Delta\mu, a, \sigma, r) = -\Delta\mu r^3 + a\sigma r^2$.

Proposed statement 2. (i) For all $\Delta\mu, a, \sigma > 0$: $r^\star > 0$; $G(r) \le G(r^\star)$ for all $r \ge 0$; and $G(r) = G(r^\star)$ with $r \ge 0$ implies $r = r^\star$. (ii) For all $\Delta\mu > 0$ and all a, σ: $G(r^\star) = \frac{1}{2}\Delta\mu\,(r^\star)^3$.

Part (i) restates canon `critical_nucleus`, and part (ii) restates `barrier_half_identity` (file sha256 `667a826a...ec28132`). Both follow from the factorisation $G(r^\star) - G(r) = \Delta\mu\,(r - r^\star)^2\,(r + r^\star/2)$. The kernel's go/no-go rule compares cluster size with $n^\star = (r^\star)^3$. Below $n^\star$ the cluster is on the uphill side of the barrier. Negative controls: $G(-1) = 5/2 > G(r^\star) = 1/2$ at $(\Delta\mu, a, \sigma) = (1, 1, 3/2)$, so the restriction $r \ge 0$ is essential. $G(r^\star) \ne \Delta\mu(r^\star)^3$, so the factor $\frac{1}{2}$ is substantive. $\Delta\mu = -1$ gives $r^\star < 0$.

```
def Statement : Prop :=
  (∀ Δμa σ: ℝ, 0 < Δμ→0 < a →0 < σ→
    0 < rStar Δμa σ∧
    (∀ r : ℝ, 0 ≤r →G Δμa σr ≤G Δμa σ(rStar Δμa σ)) ∧
    (∀ r : ℝ, 0 ≤r →G Δμa σr = G Δμa σ(rStar Δμa σ) →r = rStar Δμa σ)) ∧
  (∀ Δμa σ: ℝ, 0 < Δμ→
    G Δμa σ(rStar Δμa σ) = (1 / 2) * Δμ* (rStar Δμa σ) ^ 3)
```

5.2 `P2.Afforestation`: cubic efficiency and seeding alignment

Definitions: $\mathrm{eff}(a, \sigma, \Delta\mu, C) = C\,(3\Delta\mu/(2a\sigma))^3$ and $\eta(n, m) = \langle n, m\rangle^2/(\|n\|^2\|m\|^2)$.

Proposed statement 3. (i) For all $a, \sigma, \Delta\mu, t > 0$ and all C: $\mathrm{eff}(a, \sigma, t\Delta\mu, C) = t^3\,\mathrm{eff}(a, \sigma, \Delta\mu, C)$. (ii) In every real inner-product space E (universe-polymorphic, as in the canon), for $n, m \neq 0$: $0 \leq \eta(n, m) \leq 1$.

These restate the canon theorems `establishment_efficiency_cubic_in_drive_over_tension` and `seeding_efficiency_eq_cos2_theta` (file sha256 `3ebe33bf...9526f`).

Remark 4. The second canon theorem is named "eff $= \cos^2\Theta$". Its docstring also mentions $\eta = 1$ iff $n \parallel m$, and $1 - \eta = \sin^2\Theta$. Its Lean conclusion proves only the bound $0 \leq \eta \leq 1$, and the proposal restates only that bound. If the kernel uses the equality case, that case needs its own statement.

Sanity examples pin eff $= C/n^\star$ (27/8 at unit parameters), the factor 8 when the drive is doubled, and the factor 8 when σ is halved. They also pin $\eta = 1$ for parallel and $\eta = 0$ for orthogonal vectors in $\mathbb{R}^2$. Negative controls: doubling the drive does not double the efficiency, and the unnormalised $\langle 2, 3\rangle^2 = 36$ exceeds 1.

```
def Statement : Prop :=
  (∀ a σΔμC t : ℝ, 0 < a →0 < σ→0 < Δμ→0 < t →
    eff a σ(t * Δμ) C = t ^ 3 * eff a σΔμC) ∧
  (∀ (E : Type*) [NormedAddCommGroup E] [InnerProductSpace ℝE] (n m : E),
    n ≠0 →m ≠0 →0 ≤eta n m ∧eta n m ≤1)
```

5.3 `P2.Harmonization`: one price suffices

Definitions: $\mathrm{br}(\lambda)_i = b_i - \lambda/a_i$, $J(x) = \sum_i \frac{a_i}{2}(x_i - b_i)^2$, $\lambda^\star = (\sum_i b_i - X)/\sum_i a_i^{-1}$ and $x_i^\star = \mathrm{br}(\lambda^\star)_i$.

Proposed statement 5. For all $N > 0$, all $a, b \in \mathbb{R}^N$ with every $a_i > 0$, and all X: (i) $\sum_i \mathrm{br}(\lambda^\star)_i = X$; and (ii) every y with $\sum_i y_i = X$ satisfies $J(x^\star) \leq J(y)$.

These restate `ght_market_clears` and `ght_strong_duality_decentralized_eq_centralized` (file sha256 `cfdead65...110aec`). On the canon instance $N = 2$, $a = b = (1, 1)$, $X = 0$, the sanity examples give $\lambda^\star = 1$, $x^\star = (0, 0) \neq b$, $J(x^\star) = 1$, and $J(1, -1) = 2$. Negative controls:

- The infeasible ideal $y = b$ has $J = 0 < 1$, so the feasibility hypothesis is essential.
- The price $\lambda = 0$ does not clear.
- With $N = 0$ the aggregate is $0 \neq X = 1$, so $N > 0$ is essential.

```
def Statement : Prop :=
  ∀(N : ℕ) (a b : Fin N →ℝ), 0 < N →(∀ i, 0 < a i) →∀X : ℝ,
    (∑ i, br a b (lamStar a b X) i) = X ∧
    (∀ y : Fin N →ℝ, (∑ i, y i) = X →J a b (xStar a b X) ≤J a b y)
```

6 Results

We observe the following.

- For each proposal, a private port of the canon proof elaborated against a local build of the module. It depends only on `propext`, `Classical.choice` and `Quot.sound`. We do not present these as proofs of the proposals. The proofs belong to later tasks, against the sealed modules.

Proposal	Canon theorems restated	Canon file sha256 (prefix)	Probes closed
`P2.Restoration`	`critical_nucleus`, `barrier_half_identity`	`667a826a`	0/28
`P2.Afforestation`	`establishment_efficiency_...`, `seeding_efficiency_...`	`3ebe33bf`	0/28
`P2.Harmonization`	`ght_market_clears`, `ght_strong_duality_...`	`cfdead65`	0/28

Table 1: Proposed sealed statements. All three modules compile under the pinned toolchain, and their sanity examples elaborate. The probe column counts how many of our 28 one-shot tactics closed `Statement` locally. The verifier runs its own probes.

- Several canon hypotheses are not needed mathematically but are kept for faithfulness:
  - In Lean's convention $x/0 = 0$, the half-barrier identity also holds at $\Delta\mu = 0$.
  - The cubic law is a ring identity, so it needs none of its positivity hypotheses.
  - The alignment bound holds trivially when $n = 0$ or $m = 0$, since then $\eta = 0$.
- One canon name over-promises: the "$\cos^2\Theta$" theorem proves only the bound $0 \le \eta \le 1$.

7 What the statements settle and what they do not

- **Settled, once sealed and proved:** the internal mathematics of each kernel's rule inside its stated model.
- **Not settled:** any parameter value. This covers $\Delta\mu$, σ, a, C, the curvatures a_i, the ideals b_i and the target X. These come from field evidence and remain ASSUMED.
- **Not settled:** whether the models are adequate. The models in question are classical nucleation for tree-cluster establishment and quadratic dissipation for stewardship coordination. Nucleation rates, ecological temperature and stochastic establishment are outside the statements.
- **Not settled:** code conformance. The kernels' runtime identifiers and TypeScript sources were not among this session's inputs.

8 Limitations

The mathematics is elementary, and its value lies in being faithful and re-checkable. Faithfulness was judged by hand against the canon text. No tool checks that a proposal matches its canon theorem. Our probe list is our own guess at the verifier's and may differ from it. `P2.Afforestation` is universe-polymorphic, following the canon's `Type*`. We checked locally that a polymorphic candidate `theorem main : Statement` and the verifier's statement-match template both elaborate. A universe-monomorphic seal (`Type`) would be a mild weakening. The task brief mentions a guidance file (`STATEMENT_PROPOSAL.md`) that was not among the inputs. We followed the brief and the format of the existing sealed module instead.

9 Reproducibility

Toolchain `leanprover/lean4:v4.28.0`, Mathlib commit `8f9d9cff6bd728b17a24e163c9402775d9e6a365`. Canon repository commit `80d32904d751b45f415e23f872510cc2eddca3ff`. The canon file hashes are listed in the proposals' rationale files. Each proposal compiles with `lean proposals/P2.`*Name*`.lean`. The sealed Carbon Continuity statement has sha256 `a5a6e8c6...ffded`. To check its proof, run `viridis-lean-check proofs/P2.CarbonContinuity.lean P2.CarbonContinuity`. No automated prover jobs and no external data were used in either session.

10 Changelog

- 2026-10-29 (P2-T001): First version. Ported the canon threshold proof to the sealed statement `P2.CarbonContinuity`, and wrote the scope statement.
- 2026-10-30 (P2-T002): Proposed sealed statements `P2.Restoration`, `P2.Afforestation` and `P2.Harmonization`, with rationales, sanity examples and negative controls. Added a method section for writing statements. Condensed the Carbon Continuity material into its own section, and retitled the paper to cover the P2 program.

References

[1] Abraham Berman and Robert J. Plemmons. *Nonnegative Matrices in the Mathematical Sciences*, volume 9 of *Classics in Applied Mathematics*. SIAM, Philadelphia, PA, 1994. doi: 10.1137/1.9781611971262.

[2] Stephen Boyd and Lieven Vandenberghe. *Convex Optimization*. Cambridge University Press, Cambridge, 2004. doi: 10.1017/CBO9780511804441.

[3] Justin D. Hart. Carbon continuity after wildfire: A dynamic portfolio framework and regenerative coupling threshold for natural carbon assets. Zenodo, version 0.2.0-formal, August 2026.

[4] Dimo Kashchiev. *Nucleation: Basic Theory with Applications*. Butterworth-Heinemann, Oxford, 2000. doi: 10.1016/B978-0-7506-4682-6.X5000-8.

[5] The mathlib Community. The Lean mathematical library. In *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs*, CPP 2020, pages 367–381. ACM, 2020. doi: 10.1145/3372885.3373824.